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Final Project

Contagion or Common Factor?

Measuring Connectedness among Emerging Equity Markets

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Summary of the Base Paper and of the Network Estimators

Diebold and Yılmaz (2014) propose to measure how connected a set of assets is through the variance decomposition of an approximating VAR. The model is estimated and used to forecast each variable H periods ahead; $\theta_{ij}(H)$ is the share of the forecast-error variance of variable i due to shocks in variable j. Because shocks are usually correlated, the authors use the generalized variance decomposition of Koop et al. (1996) and Pesaran and Shin (1998), whose results do not depend on the ordering of the variables; its rows no longer sum to one, so they are normalized. The resulting matrix, augmented with its row and column sums, is the *connectedness table*. Each off-diagonal entry measures how much j affects i; the off-diagonal row sum gives how much a variable receives from the others ("from"), the column sum how much it transmits to them ("to"), and their difference whether it is a net transmitter or a net receiver. The average of all off-diagonal entries is the total connectedness of the system. Re-estimating the model over a rolling window shows how connectedness changes over time.

The main conceptual contribution of the paper is that the connectedness table is a weighted, directed network: "from" and "to" connectedness are the in- and out-degrees of each node, and total connectedness is the mean degree. The two directions also correspond to known systemic-risk measures: what a firm receives resembles exposure measures such as marginal expected shortfall, and what it transmits resembles contribution measures such as ΔCoVaR.

In the application, the authors study the daily realized volatilities of 13 large US financial institutions from May 1999 to April 2010, with a VAR(3), a 12-day horizon and a 100-day rolling window. Total connectedness is 78.3% in the full sample. Connectedness "from" others is tightly distributed (70–82%), whereas connectedness "to" others ranges from 53% to 106%, with Citigroup as the largest net transmitter. Over time the index shows two large cycles, around 2000–2003 and during the 2007–2009 crisis, when it reaches its maximum of 89.2% after the Lehman bankruptcy; the results are robust to the window, the horizon and the ordering of the variables. Our application changes the set of assets (emerging-market country funds), the object (returns instead of volatilities) and the frequency (weekly instead of daily).

Three limitations of the approach are relevant for our work. First, the VAR has N^2p coefficients and is often estimated with few observations. Second, every pair of variables ends up connected, even with a very small weight, so the network never has zero links. Third, the variance decomposition cannot tell whether assets move together because they affect each other or because they are all hit by the same shock. The two estimators below address these limitations.

NETS (Barigozzi and Brownlees, 2019) models the panel as a VAR in which both the autoregressive matrices and the precision (or concentration) matrix of the innovations (the inverse of their covariance matrix) are sparse, that is, most of their elements are zero. The dependence structure is then represented by networks. The Granger network is directed: there is a link from j to i when the coefficient on the lags of j in the equation for i is different from zero, so that j has predictive power for i. The contemporaneous network is undirected and is built from the partial correlations of the VAR innovations, which measure the relationship between two variables in the same period after controlling for all the others. The two combine into a long-run partial-correlation network. Estimation is carried out jointly by a lasso-type procedure that sets irrelevant links to zero, and, unlike the Diebold–Yılmaz approach, it requires no identifying assumptions on the shocks. In their application to the volatilities of 90 large US stocks, the authors find that the network captures part of the variability not explained by common volatility factors and improves out-of-sample forecasts. Its main limitation is that it does not model common factors explicitly: when a factor affects the whole panel, the dependence between series is no longer sparse and the central assumption of the model fails.

FNETS (Barigozzi et al., 2024) addresses this limitation with a factor-adjusted VAR. Each series is decomposed into a common component, driven by a small number of factors in a generalized dynamic factor model (Forni et al., 2000) and estimated by dynamic principal components, so that factors can affect each series with different lags; and an idiosyncratic component, modeled

as a sparse VAR. Since the idiosyncratic component is not observed, its parameters are estimated by an ℓ_1-regularized Yule–Walker estimator built from the idiosyncratic autocovariances implied by the estimated spectral density. The same three networks as in NETS are then estimated, and they reflect only the dependence that cannot be attributed to the common factors. The method also allows the common and idiosyncratic components to be forecast separately. The authors prove the consistency of the estimators as both the number of series and the sample size grow, also under heavier-than-Gaussian tails, and implement the procedure in the R package `fnets` (Owens et al., 2024). For our purpose, FNETS is the key method, because it separates the part of the connectedness among emerging markets that is due to global factors from the part that reflects links between the markets themselves.

1 Motivation

Emerging equity markets are exposed to two kinds of shocks. Some are global: changes in US monetary policy, in the dollar or in investors' appetite for risk move all of them at once, a pattern that Rey (2013) describes as a global financial cycle. Others are local, such as an election, a currency crisis or a change in economic policy in one country, and the question is whether, and through which countries, they spread to the rest. Telling the two apart matters for several reasons. For investors, the benefit of diversifying across emerging markets depends on how much of their co-movement is a common component that no portfolio of these markets can diversify away. For policy makers, a high degree of connectedness calls for different responses if it reflects bilateral transmission (contagion) or a common exposure to the same external shock. For the empirical literature, the distinction is an old one: Forbes and Rigobon (2002) showed that much apparent contagion during crises is high interdependence driven by common shocks and higher volatility.

The tools to measure connectedness have multiplied, and they do not all measure the same thing. The variance-decomposition approach of Diebold and Yılmaz (2014) summarizes how much of each market's forecast-error variance is due to shocks elsewhere; it is dense by construction and does not separate common from bilateral transmission. Sparse network estimators such as NETS (Barigozzi and Brownlees, 2019) keep only the links supported by the data, and FNETS (Barigozzi et al., 2024) first removes the common factor and then estimates the network of what is left. Applied to the same data, these methods can give quite different pictures of how connected markets are, who transmits and who receives, and how stable the network is. Before using a connectedness measure to draw conclusions about contagion, it is therefore worth knowing how much of the measured connectedness depends on the estimation method.

Our sample of 17 emerging markets offers clear examples of both kinds of shock. In the week of the Argentine primary elections of August 2019 the Argentine fund (ARGT) lost 28.4% (in log terms), its largest weekly fall in the sample: a large, country-specific shock. Seven months later, the COVID-19 crash of March 2020 produced the largest weekly losses of almost every market at the same time: 14 of the 17 markets recorded their worst week in March 2020. The Argentine episode illustrates the question we ask: whether a large local shock spreads to other markets, and whether the methods agree on that.

2 Introduction

This paper asks how measured connectedness among emerging equity markets changes with the method used to estimate it, and what share of it reflects a global common factor rather than idiosyncratic links between markets. We study weekly returns of US-listed country funds for 17 emerging markets in Latin America, Asia and EMEA, together with the S&P 500 as a global reference, from March 2011 to October 2026, and we estimate connectedness in four ways: the variance-decomposition measures of Diebold and Yılmaz (2014) on a VAR estimated by OLS;

the same measures on a sparse VAR estimated by the lasso with time-series cross-validation; NETS (Barigozzi and Brownlees, 2019), which estimates sparse Granger, contemporaneous and long-run partial-correlation networks; and FNETS (Barigozzi et al., 2024), which estimates the same networks after removing the common factor. All methods are estimated on the same rolling windows and mapped into comparable measures, and we compare them on three criteria: the density of the networks, the agreement between methods, and their stability over time.

We test three hypotheses. First, regularization alone should matter little for total connectedness at the weekly frequency, because the cross-market dependence of returns is mostly contemporaneous rather than lagged. Second, a substantial share of the connectedness measured by Diebold–Yılmaz should be due to a common factor, so that it falls markedly once the factor is removed, and its time variation should follow the weight of that factor. Third, sparse estimators should produce networks that are less dense, and also less stable from one window to the next, than the dense variance-decomposition network.

The results support the first two hypotheses and only partly the third. The lasso VAR sets almost all lag coefficients to zero yet gives nearly the same connectedness as Diebold–Yılmaz (80.0% against 81.0% on average over windows, correlation 0.98). NETS gives a lower level (61.5%) and keeps about half of the possible long-run links, while tracking the Diebold–Yılmaz series closely (correlation 0.89). FNETS finds a single common factor in every window, which explains on average 58% of the variance of the returns. Measured on the same scale as Diebold–Yılmaz, spillovers between the idiosyncratic components account on average for only 19.6% of the variance of a market, against the 81.0% that Diebold–Yılmaz attributes to shocks in other markets, and a benchmark with the same common factor and no links between markets already produces 81.1% connectedness. Equivalently, total connectedness falls by 41% when the factor is removed, a share that is stable across window lengths, horizons, the exclusion of the S&P 500 and the exclusion of 2020, although it changes when two factors are imposed. Over time, Diebold–Yılmaz connectedness and the weight of the common factor move together. The sparse networks are indeed much less dense, but stability depends on the measure: NETS connectedness jumps more from one window to the next than Diebold–Yılmaz, yet its strongest long-run links are more persistent; the idiosyncratic network of FNETS is both the sparsest and the least stable. South Africa, Korea and Taiwan are the main net transmitters and Turkey, Vietnam and Argentina the main net receivers in the dense and NETS systems; once the common factor is removed, Argentina is close to neutral.

Our contribution is to put the base paper's connectedness measures side by side with sparse and factor-adjusted network estimators on the same data, windows and metrics, for a set of emerging markets, and to quantify the part of measured connectedness that is a common factor. Section 3 describes the data, Section 4 the methods and comparison criteria, Section 5 the results, and Sections 6 and 7 discuss them and conclude. Robustness checks, country-level results and bootstrap intervals are in the appendix.

3 Data

Universe and sample. We use US-listed, USD-denominated country exchange-traded funds (ETFs) as tradable proxies for 17 emerging equity markets: Argentina (ARGT), Chile (ECH), Brazil (EWZ), Mexico (EWW) and Peru (EPU) in Latin America; South Korea (EWY), Taiwan (EWT), India (EPI), China (FXI), Thailand (THD), Indonesia (EIDO), Malaysia (EWM), Vietnam (VNM) and the Philippines (EPHE) in Asia; South Africa (EZA), Turkey (TUR) and Poland (EPOL) in EMEA. The S&P 500 ETF (SPY) is included as a global reference node. Funds were selected if their price history starts by 3 March 2011 (7-day tolerance), at most 5% of weeks are missing, and average daily traded volume is at least USD 3 million. Colombian funds (ICOL, GXG), INDA and MCHI did not pass the criteria; India and China are represented by EPI and FXI. The common sample runs from 11 March 2011 to 23 October 2026, i.e. 816 weeks,

with no missing values.

Returns and frequency. Prices are dividend- and split-adjusted, so returns are total returns. We use weekly Friday-to-Friday log returns. Choosing the weekly frequency is a decision of ours, for three reasons: weekly returns retain some cross-market predictability, which is the dynamic dependence a VAR-based network is meant to capture; weekly aggregation mitigates the non-synchronous trading between Asian and EMEA local markets and their US-listed ETFs; and it reduces the microstructure noise of daily ETF prices. We also choose to study returns rather than volatilities. Diebold and Yılmaz (2014) measure the connectedness of realized volatilities; with daily prices the same could be done here, by building weekly realized volatilities, but we leave volatility connectedness as an extension and focus on the transmission of return shocks. Returns are not devoid of cross-market predictive content: in a full-sample VAR(1) on the 18 series, 59 of the 324 slope coefficients (18.2%) are significant at the 5% level, against 5% expected by chance. Our networks therefore describe the transmission of return shocks, not of risk.

Descriptive statistics. Table 1 reports summary statistics. All series are strongly leptokurtic and mostly negatively skewed; an augmented Dickey–Fuller test rejects a unit root for every series. Five weekly log returns exceed 25% in absolute value (ARGT in the week of the August 2019 primary elections and four observations in March 2020); they are genuine market moves and are kept. Argentina illustrates how much these markets are driven by local politics: it is the third most volatile market (weekly standard deviation 4.2%, after Turkey and Brazil), and both its largest weekly gains, 24.0% in the week after the October 2025 legislative elections and 15.1% after the November 2023 presidential runoff, and its largest loss, after the August 2019 primaries, are electoral weeks. The March 2020 crash dominates the tails: of the ten largest weekly drops pooled across the 17 emerging markets, eight occur in the weeks ending 13 and 20 March 2020, and 14 of the 17 markets record their worst week in March 2020.

Table 1. Descriptive statistics of weekly log returns, 2011-03-11 to 2026-10-23 (816 weeks). Kurtosis is not in excess.

Ticker	Country	Region	Start	Mean (%)	Std (%)	Min (%)	Max (%)	Skew.	Kurt.
ARGT	Argentina	Latin America	2011-03-11	0.15	4.18	-28.4	24.0	-0.51	8.9
ECH	Chile	Latin America	2011-03-11	-0.03	3.56	-21.2	16.2	-0.36	5.6
EWZ	Brazil	Latin America	2011-03-11	0.01	4.26	-26.5	22.2	-0.18	6.4
EWW	Mexico	Latin America	2011-03-11	0.07	3.29	-24.0	10.3	-0.74	7.8
EPU	Peru	Latin America	2011-03-11	0.12	3.35	-19.3	13.3	-0.39	5.9
EWY	South Korea	Asia	2011-03-11	0.14	3.66	-20.3	16.0	-0.40	6.6
EWT	Taiwan	Asia	2011-03-11	0.23	2.85	-12.6	11.2	-0.43	4.4
EPI	India	Asia	2011-03-11	0.09	2.90	-18.8	13.3	-0.35	6.3
FXI	China	Asia	2011-03-11	0.01	3.47	-12.6	17.0	0.03	4.5
THD	Thailand	Asia	2011-03-11	0.07	2.91	-23.5	9.6	-0.85	9.0
EIDO	Indonesia	Asia	2011-03-11	-0.08	3.68	-33.6	20.7	-1.25	14.6
EWM	Malaysia	Asia	2011-03-11	0.01	2.25	-12.2	10.9	-0.22	6.5
VNM	Vietnam	Asia	2011-03-11	-0.01	3.15	-19.3	11.9	-0.35	5.9
EPHE	Philippines	Asia	2011-03-11	0.02	2.92	-19.3	14.4	-0.57	7.4
EZA	South Africa	EMEA	2011-03-11	0.07	4.06	-26.9	14.1	-0.57	6.5
TUR	Turkey	EMEA	2011-03-11	-0.03	4.60	-23.0	20.0	-0.63	6.1
EPOL	Poland	EMEA	2011-03-11	0.08	3.66	-25.1	15.2	-0.58	6.5
SPY	United States	Global	2011-03-11	0.25	2.23	-15.7	11.4	-0.74	9.3

Limitations. (i) ETF returns embed exchange-rate movements against the USD and are subject to non-synchronous trading. (ii) FXI tracks the FTSE China 50 Index, i.e. 50 large-cap Chinese companies listed in Hong Kong, and EPI tracks the earnings-weighted WisdomTree India Earnings Index, so neither is a broad market index of its country. (iii) Colombia is absent. (iv) Liquidity requirements and the common start date exclude younger funds. (v) Volatility connectedness, the object of the base paper, is not studied here.

4 Methods

Let $y_t \in \mathbb{R}^N$ be the vector of weekly returns ($N = 18$). All methods are estimated on rolling windows of $w = 104$ weeks; each estimate is dated at the last week of its window. For the comparison, all methods use the same 179 windows (step of four weeks, first window ending 1 March 2013).

4.1 Diebold–Yılmaz connectedness

We fit a VAR(p), $y_t = c + \sum_{k=1}^{p} A_k y_{t-k} + \varepsilon_t$, $\mathbb{E}[\varepsilon_t \varepsilon_t'] = \Sigma$, by OLS, with $p \in \{1, 2\}$ selected by BIC in each window. The cap $p \leq 2$ is an adaptation: with $N = 18$ and about 100 observations per window, $p = 3$ would mean 55 regressors per equation. With MA coefficients Ψ_h ($\Psi_0 = I$), the H-step generalized variance share (Pesaran and Shin, 1998) is

$$\theta_{ij}(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (e_i' \Psi_h \Sigma e_j)^2}{\sum_{h=0}^{H-1} e_i' \Psi_h \Sigma \Psi_h' e_i}, \qquad \tilde{\theta}_{ij} = \theta_{ij} \Big/ \sum_j \theta_{ij}, \tag{1}$$

which is invariant to the ordering of the variables. Following Diebold and Yılmaz (2014), $C_i^{\text{from}} = 100 \sum_{j \neq i} \tilde{\theta}_{ij}$, $C_j^{\text{to}} = 100 \sum_{i \neq j} \tilde{\theta}_{ij}$, net connectedness $C_i^{\text{net}} = C_i^{\text{to}} - C_i^{\text{from}}$, pairwise net $100(\tilde{\theta}_{ji} - \tilde{\theta}_{ij})$ (positive when i transmits more to j than it receives from j), and total connectedness $C = \frac{100}{N} \sum_{i \neq j} \tilde{\theta}_{ij}$. The baseline horizon is $H = 4$ weeks. The base paper uses daily data, 100-day windows and $H = 12$ days; we keep the logic (a horizon of about one month, a window long enough for a stable VAR) on the weekly scale.

4.2 Sparse VAR (lasso)

As a bridge between the dense DY VAR and the sparse network estimators, we estimate each VAR equation by the lasso (Tibshirani, 1996) on lags $1, \ldots, 4$ (standardized regressors, unpenalized intercept), so that the penalty performs lag selection. The penalty is chosen per equation by time-series cross-validation (five expanding folds, test blocks always after the training block; random K-fold would leak future information), over 40 log-spaced values; the extended BIC (Chen and Chen, 2008) is an alternative. The same GFEVD and connectedness measures are computed on the lasso VAR, so differences with DY isolate the effect of regularization.

4.3 NETS

We estimate NETS with one lag, i.e. a sparse VAR(1) and a sparse precision (concentration) matrix $K = \Sigma^{-1}$ of the innovations, using adaptive lasso weights (Zou, 2006; Peng et al., 2009). Since the `nets` package requires the penalty to be supplied, we select it by BIC over 15 log-spaced values. The Granger network is given by the non-zero entries of A_1, the contemporaneous network by the partial correlations $-K_{ij}/\sqrt{K_{ii}K_{jj}}$, and the long-run network by the partial correlations of $K_{LR} = (I - A(1))' K (I - A(1))$. Each window is standardized before estimation.

4.4 FNETS

Writing $y_t = \chi_t + \xi_t$ for the common and idiosyncratic components, we use the unrestricted (generalized dynamic) factor model, select q with the Hallin and Liška (2007) information criterion (criterion IC5, the package default), estimate the idiosyncratic VAR(1) by lasso with the penalty chosen by the package's BIC, and estimate contemporaneous and long-run partial correlations of ξ_t with the package's CLIME-type estimator (Cai et al., 2016) and adaptive thresholding. The share of variance explained by the common component is $\text{var}(\chi_{it})/\text{var}(y_{it})$. Applying the GFEVD to the idiosyncratic VAR yields *idiosyncratic connectedness*. The generalized variance decomposition uses the estimated idiosyncratic innovation covariance, after the positive-definite

adjustment described below, and not the thresholded partial-correlation network, which is used only to describe the network structure; idiosyncratic connectedness can therefore be positive even when the thresholded network has no links, as in the full sample (Section 5.1). Comparing it with DY measures how much connectedness is due to the common factor. With $N = 18$ and $T = 104$ the estimated idiosyncratic innovation covariance is often not positive definite; we project it onto the positive-definite cone by clipping eigenvalues at 1% of the mean absolute eigenvalue, and clip partial correlations to $[-1, 1]$ (the support of the network is unchanged). Both adjustments, and the fact that the idiosyncratic partial-correlation network can be empty in long samples (Section 5), indicate that FNETS operates here at the edge of its intended setting of large N and T; we treat its idiosyncratic networks with corresponding caution.

4.5 Comparable measures and comparison criteria

The lag orders differ across methods ($p \leq 2$ selected by BIC for DY, up to four lags for the lasso VAR, one lag for NETS and FNETS), following the standard specification of each estimator; since lagged dependence is negligible at the weekly frequency (Section 5), this does not affect the comparison. Every method is mapped to the same objects: a VAR on which the same GFEVD connectedness is computed; a directed Granger network (sum over lags of the absolute coefficients); and contemporaneous and long-run partial-correlation networks (for DY and the lasso VAR, implied by the estimated VAR). We compare methods on three criteria:

1. *Density*: share of non-zero links in each network; for the dense GFEVD table, share of pairwise shares above $1/N$; and total connectedness.

2. *Agreement*: correlation of total and net connectedness series across methods; per-window Spearman correlation between the off-diagonal entries of two methods' matrices; Jaccard similarity of their strongest 10% of links.

3. *Stability*: variability of total connectedness, and Spearman correlation and top-link Jaccard similarity between consecutive windows' matrices.

Finally, the share of DY connectedness that disappears once the common factor is removed is $1 - C^{\text{FNETS}}/C^{\text{DY}}$, averaged over windows.

Because DY connectedness is a share of the variance of the returns while FNETS connectedness is a share of the variance of the idiosyncratic component only, we also express the role of the factor on the scale of total variance. For market i in window w, the share of its total variance due to idiosyncratic spillovers is $IS_{i,w} = (1 - s_{i,w})\, C_{i,w}^{\text{from,FNETS}}$, where $s_{i,w} = \text{var}(\chi_{it})/\,\text{var}(y_{it})$ is the common share; we average it over markets and windows and compare it with DY total connectedness, which is the average of C_i^{from}. We also compute a pure-factor benchmark: in each window we build $\Sigma^B = \text{cov}(\hat{\chi}_t) + \text{diag}\,(\,\text{var}(\hat{\xi}_{1t}), \ldots, \text{var}(\hat{\xi}_{Nt}))$, which keeps the estimated common component and sets all idiosyncratic covariances and lags to zero, and apply the generalized variance decomposition to it (with no lags, $\theta_{ij} = (\Sigma_{ij}^B)^2/(\Sigma_{ii}^B \Sigma_{jj}^B)$). Its total connectedness C^B is the connectedness that a common factor would produce if there were no links between markets.

4.6 Robustness

We rerun all methods with windows of 78 and 130 weeks, horizons of 2 and 8 weeks, without SPY, and excluding 2020; we also rerun NETS with a finer grid of 60 penalty values and FNETS with the number of factors fixed at $q = 2$. For the exclusion of 2020, every rolling window containing any week of 2020 is dropped (the resulting gap is not interpolated), and static full-period connectedness is estimated separately for 2011–2019 and 2021–2026.

For the main full-sample quantities (DY connectedness, FNETS idiosyncratic connectedness, the common share, the idiosyncratic spillover share and the pure-factor benchmark) we compute

95% confidence intervals by a moving-block bootstrap with blocks of 8 weeks and 200 bootstrap samples, re-estimating every step of each method, including lag and factor-number selection, in each sample (Table 6).

5 Results

5.1 Full-sample networks

Table 4 (appendix) reports static connectedness estimated once on the whole sample; these figures differ from the averages over rolling windows reported in Table 2 and below (e.g. 81.9% against 81.0% for Diebold–Yılmaz, 34.2% against 47.2% for FNETS), because a single long-sample estimate is not the mean of short-window estimates. The gap is much larger for FNETS (13 points) than for Diebold–Yılmaz (under one point), which suggests that in short windows estimation noise adds spurious idiosyncratic dependence; if so, the rolling-window averages overstate idiosyncratic connectedness and understate the role of the common factor. The two dense estimators give almost the same total connectedness (Diebold–Yılmaz 81.9%, lasso VAR 82.1%), NETS gives 73.5%, and the idiosyncratic component identified by FNETS gives only 34.2%. The heatmap of Figure 3(a) shows that Diebold–Yılmaz connectedness is spread widely across all pairs, with only mild regional clustering: the strongest pairs are Korea–Taiwan and the Indonesia–Malaysia–Philippines–Thailand block, and South Africa is a common source of shocks for almost every market. Once the common factor is removed (Figure 3(b)), the remaining pairwise shares are concentrated in a few pairs (Korea–Taiwan above all), and the full-sample idiosyncratic partial-correlation network is empty (Figure 4, right panel). Idiosyncratic connectedness is nonetheless 34.2% because, as explained in Section 4.4, it is computed from the estimated innovation covariance and not from the thresholded network. This can mean either that, once the common factor is removed, there is little network structure left among these markets, or that the thresholded estimator is too strict to detect it with the available data; in rolling 104-week windows the idiosyncratic network is sparse (7% of possible long-run links on average) but not empty. We cannot distinguish the two explanations and list this among the limitations, together with the positive-definite adjustment described in Section 4.4. NETS retains 92 long-run partial-correlation links (60% of all possible pairs). The lagged (Granger) dependence is very weak in all sparse estimators: lasso, NETS and FNETS keep only 2.5%, 1.7% and 0.3% of the possible directed lag links in the rolling windows, so connectedness at the weekly frequency is overwhelmingly contemporaneous. This is consistent with the full-sample VAR(1) of Section 3, in which 18.2% of the slope coefficients are significant: the lagged dependence is real but weak, and in 104-week windows it does not survive the penalty.

5.2 Connectedness over time

Figure 1 shows the rolling total connectedness of each method. Diebold–Yılmaz connectedness moves between about 72% and 89%; the lasso VAR tracks it almost exactly (correlation 0.98). Because each estimate is dated at the end of its 104-week window, an episode affects the series for two years after it occurs. Connectedness is high at the start of the sample (88%), falls to about 76% in 2014–2015, rises to about 87% in 2016–2017, is around 80% in 2018–2019, and reaches its maximum (about 89%) for windows that contain the March 2020 crash, falling back to the mid-70s after those weeks leave the window (2022–2023). NETS follows the same low-frequency pattern at a lower level, mostly between 46% and 78% (5th and 95th percentiles), but is much noisier from one window to the next and drops to about 32% in a few windows. FNETS idiosyncratic connectedness is flatter, mostly between 36% and 53%, with occasional one-window drops to about 31% (15 of 179 windows fall below 40%; see the discussion of stability below), and it does not reproduce the 2020 peak. Figure 2 helps to read this: the share of variance that FNETS attributes to the common component, between 47% and 75%, moves together with

Diebold–Yılmaz total connectedness (correlation 0.95). Part of this co-movement is mechanical, since both measures increase when the correlations among all series increase, so we read it as evidence that the two move together and not as a causal statement.

5.3 Net transmitters and receivers

Figure 5 reports average net connectedness by country. In the Diebold–Yılmaz and lasso systems the main net transmitters are South Africa (+22 percentage points), South Korea (+16) and Taiwan (+12), and the main net receivers are Turkey (−22), Vietnam (−21) and Argentina (−10). NETS gives the same ranking at about half the magnitude (South Africa +10, Korea +9, Taiwan +8; Vietnam −11, Turkey −11, Argentina −5). The ranking of the countries' average net position is almost identical across DY, lasso and NETS (Spearman correlation between 0.96 and 0.99) but differs for FNETS (0.30 with DY). In the idiosyncratic network Taiwan (+9) and Indonesia (+8) are the main transmitters, Turkey is still the main receiver (−6), and Argentina is close to neutral (+1.6). Its average common share is 48.6%, the sixteenth highest of the 18 series (Table 5), so its role as a net receiver in the dense systems does not reflect a stronger exposure to the common factor: as discussed in Section 6, it is largely a consequence of the row normalization, which makes the markets least correlated with the rest appear as receivers. Indeed, the average correlation of Turkey, Vietnam and Argentina with the other markets (0.39, 0.41 and 0.47) is the lowest in the sample, against 0.61 for South Africa. The S&P 500 (SPY), included as a global reference, has an average net connectedness of +2.7 pp in the Diebold–Yılmaz system and +2.4 pp in the idiosyncratic network. These directional results call for some caution. All ETFs trade in New York, while the Asian markets (and, for part of the day, the South African and Polish markets) are already closed; a US-listed fund can therefore react to news before its local market does, and the timing can shift the estimated direction of spillovers. Weekly returns reduce this effect but do not remove it, so the transmitter role of Korea, Taiwan and South Africa may partly reflect trading hours.

5.4 Comparison of methods

Table 2 summarizes the three criteria.

Density. The Diebold–Yılmaz VAR is dense by construction. The sparse estimators remove almost all lagged links, and NETS and FNETS also sparsify the contemporaneous structure: the share of non-zero long-run partial correlations is 100% for DY and lasso (which are implied by a full innovation covariance), 52% for NETS and 7% for FNETS. Among the generalized variance shares themselves, 35% (DY) and 34% (lasso) exceed the equal-share benchmark $1/N$, against 16% for both NETS and FNETS.

Agreement. Total connectedness is highly correlated between DY and the lasso VAR (0.98) and between DY and NETS (0.89). The per-window Spearman correlation of the variance-decomposition entries with DY is 0.93 for lasso and 0.80 for NETS, and the overlap of the strongest 10% of links (Jaccard) is 0.61 and 0.51. The comparison with FNETS is of a different kind: FNETS measures connectedness of the idiosyncratic component only, after the common factor that drives most of the co-movement has been removed, whereas the other three methods measure connectedness of the returns themselves. The low overlap of their strongest links (Jaccard 0.16) should therefore not be read as disagreement about the same object. The correlation close to zero between the FNETS and DY series (0.00; 0.04 and 0.08 with lasso and NETS) is not implied by construction, however: it is an empirical result, and an informative one. It means that the time variation of Diebold–Yılmaz connectedness, which follows the weight of the common factor (Section 5.2), carries no information about the idiosyncratic network. Agreement is lower for the networks that are sparse by design: for long-run partial correlations the Spearman

Table 2. Comparison criteria across methods: averages over 179 rolling windows ($w = 104$, $H = 4$), not full-sample estimates (those are in Table 4). Densities in %; agreement measured against DY; stability is the mean Spearman correlation (variance decomposition) and top-10% Jaccard similarity (long-run partial correlations) between consecutive windows.

Method	Total conn. (%)	Granger dens. (%)	LR p.corr. dens. (%)	Corr. total w/ DY	Spearman GFEVD w/ DY	Stab. GFEVD	Stab. LRPC top
DY	81.0	100.0	100.0	–	–	0.97	0.61
Lasso VAR	80.0	2.5	100.0	0.98	0.93	0.97	0.64
NETS	61.5	1.7	51.9	0.89	0.80	0.93	0.72
FNETS (idio.)	47.2	0.3	7.3	0.00	-0.06	0.82	0.12

correlation with DY is 0.33 (lasso) and 0.36 (NETS), while lasso and NETS agree more with each other (0.66). The Granger networks barely agree (Spearman at most 0.23), as expected given how few lag links survive.

Stability. The standard deviation of total connectedness across windows is 5.1 (DY), 5.7 (lasso), 11.0 (NETS) and 4.6 (FNETS), but its mean absolute change from one window to the next is 0.7, 0.8, 4.6 and 3.0: the dense estimators move smoothly, while NETS and FNETS jump. The correlation between consecutive windows' variance-decomposition matrices is 0.97 (DY), 0.97 (lasso), 0.93 (NETS) and 0.82 (FNETS). For long-run partial-correlation links the overlap of the strongest 10% between consecutive windows is 0.61, 0.64, 0.72 and 0.12, so NETS has the most persistent sparse structure and the FNETS idiosyncratic network is the least stable. The sharp one-window drops of the FNETS series coincide with windows in which the penalized idiosyncratic VAR retains several coefficients (correlation of -0.72 between total connectedness and the number of non-zero coefficients); we therefore read them as estimation noise in short windows and not as events.

5.5 Common factor versus idiosyncratic connectedness

FNETS selects one common factor in every window, which explains on average 58% of the variance of the series. Total connectedness of the idiosyncratic VAR is 41% lower than that of the DY VAR on average (median 42%; between 26% and 61% across windows). Expressed as a share of the total variance of each market (Section 4.5), idiosyncratic spillovers average 19.6% (between 10.0% and 28.1% across windows), against 81.0% for Diebold–Yılmaz, while the pure-factor benchmark averages 81.1% and has a correlation of 0.98 with Diebold–Yılmaz connectedness over time. The loss is stable across specifications (Table 3): 38% with 78-week windows, 43% with 130-week windows, 42% without SPY and 41% when windows that touch 2020 are removed, and it does not depend on the horizon. It is larger when two factors are imposed (52%, Appendix A). In the subsamples (Table 4), DY connectedness is 83.0% in 2011–2019 and 73.5% in 2021–2026, NETS falls from 69.6% to 56.0%, while idiosyncratic FNETS connectedness rises slightly (34.1% to 39.6%) as the common-factor variance share falls from 58% to 47%. A large part of the variation of DY connectedness over time is therefore variation in the weight of the common factor.

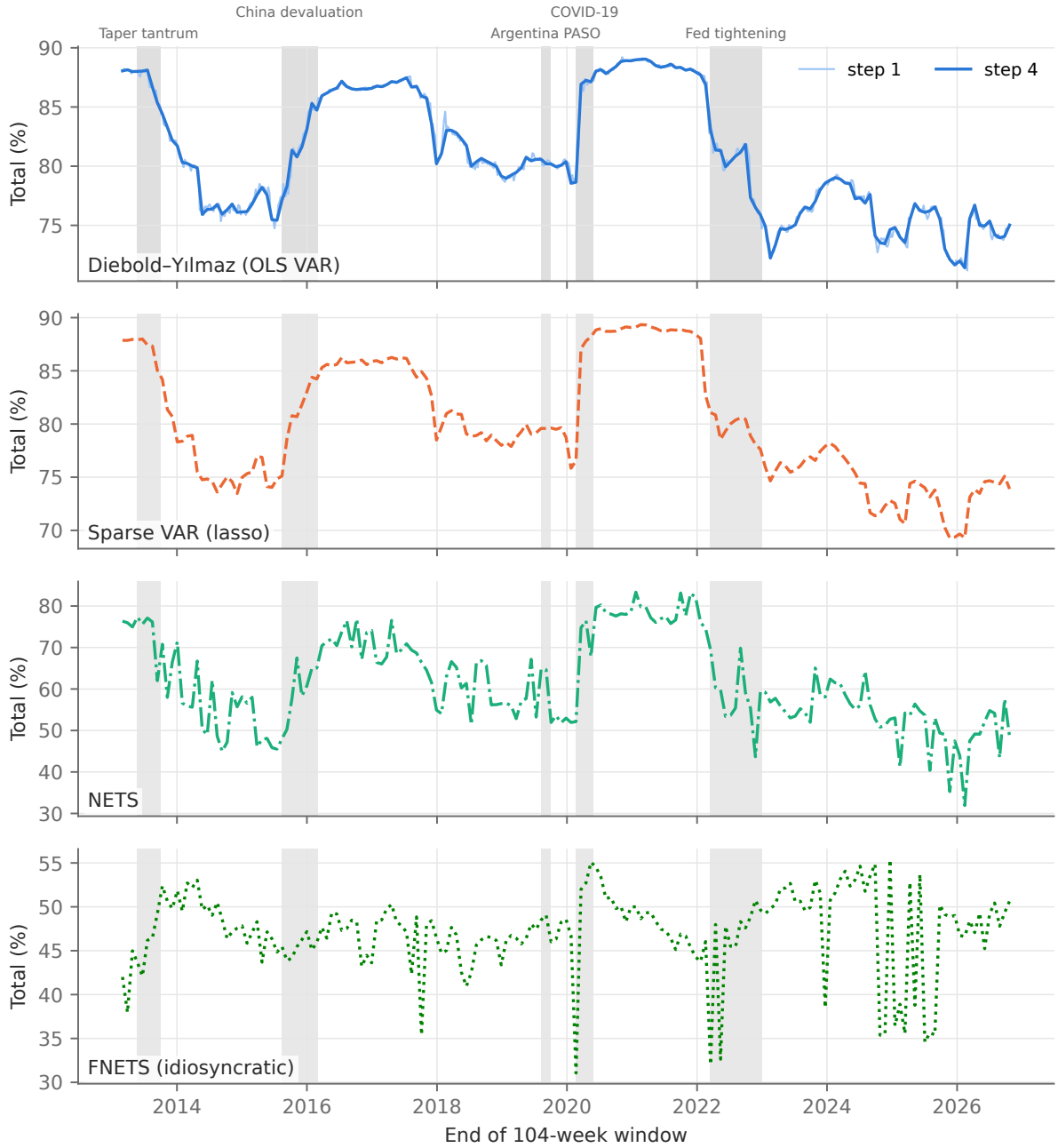


Fig. 1. Total connectedness over time by method (104-week rolling windows, $H = 4$, step 4). Estimates are dated at the end of the window. Shaded areas mark events.

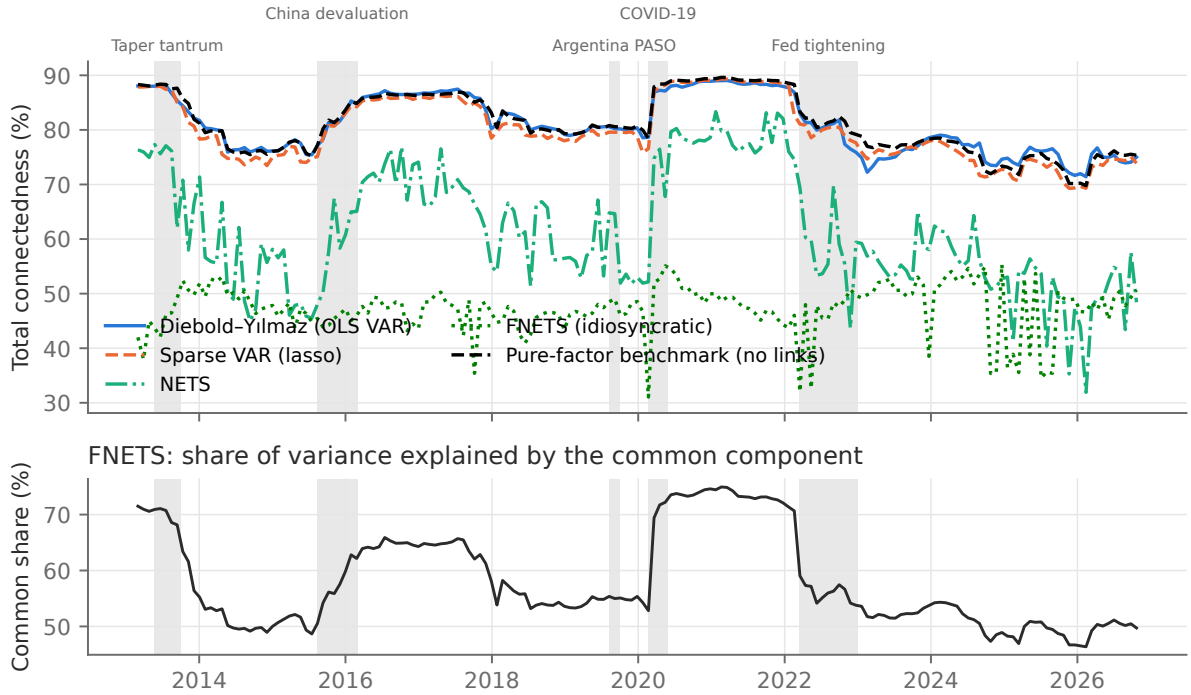
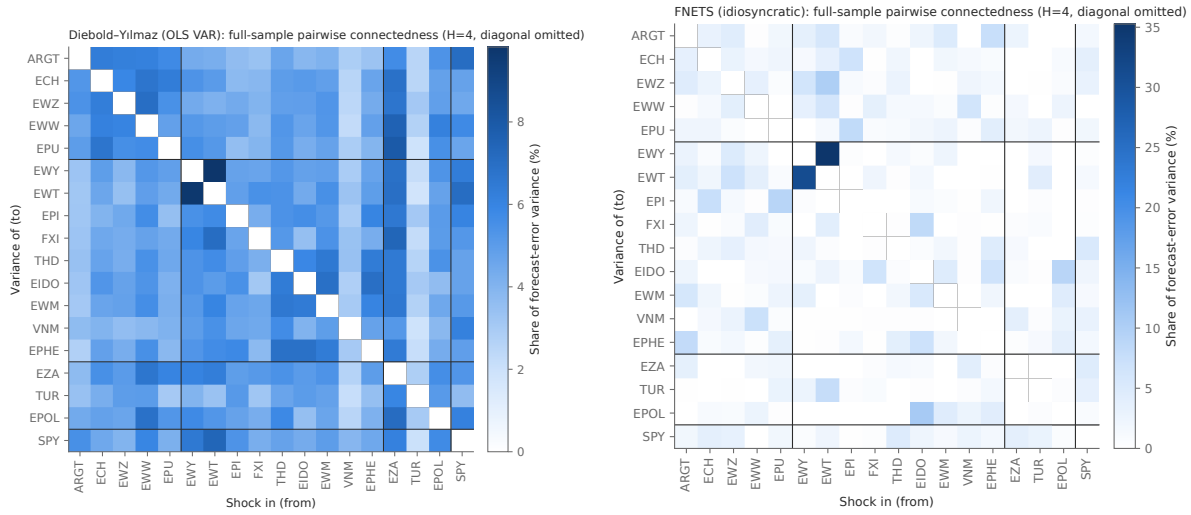


Fig. 2. Top: total connectedness of all methods on a common scale. The dashed line is the pure-factor benchmark of Section 4.5. Bottom: FNETS share of variance explained by the common component.



(a) DY (b) FNETS (idiosyncratic)

Fig. 3. Full-sample pairwise connectedness, countries grouped by region (Latin America, Asia, EMEA, global).

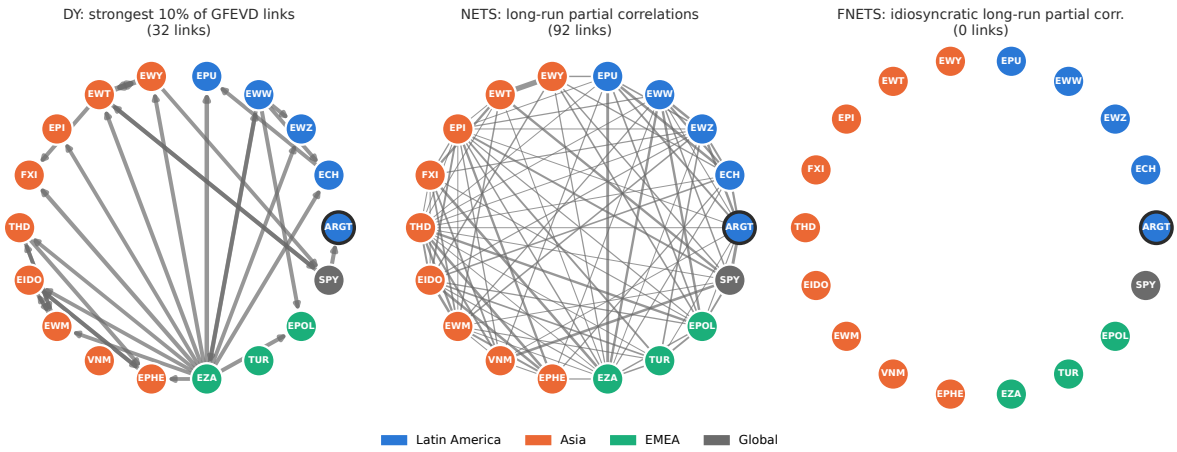


Fig. 4. Full-sample networks. Node colors denote regions; ARGT is outlined. Left: strongest 10% of Diebold–Yılmaz links. Center: NETS long-run partial correlations. Right: FNETS idiosyncratic long-run partial correlations (no links in the full sample).

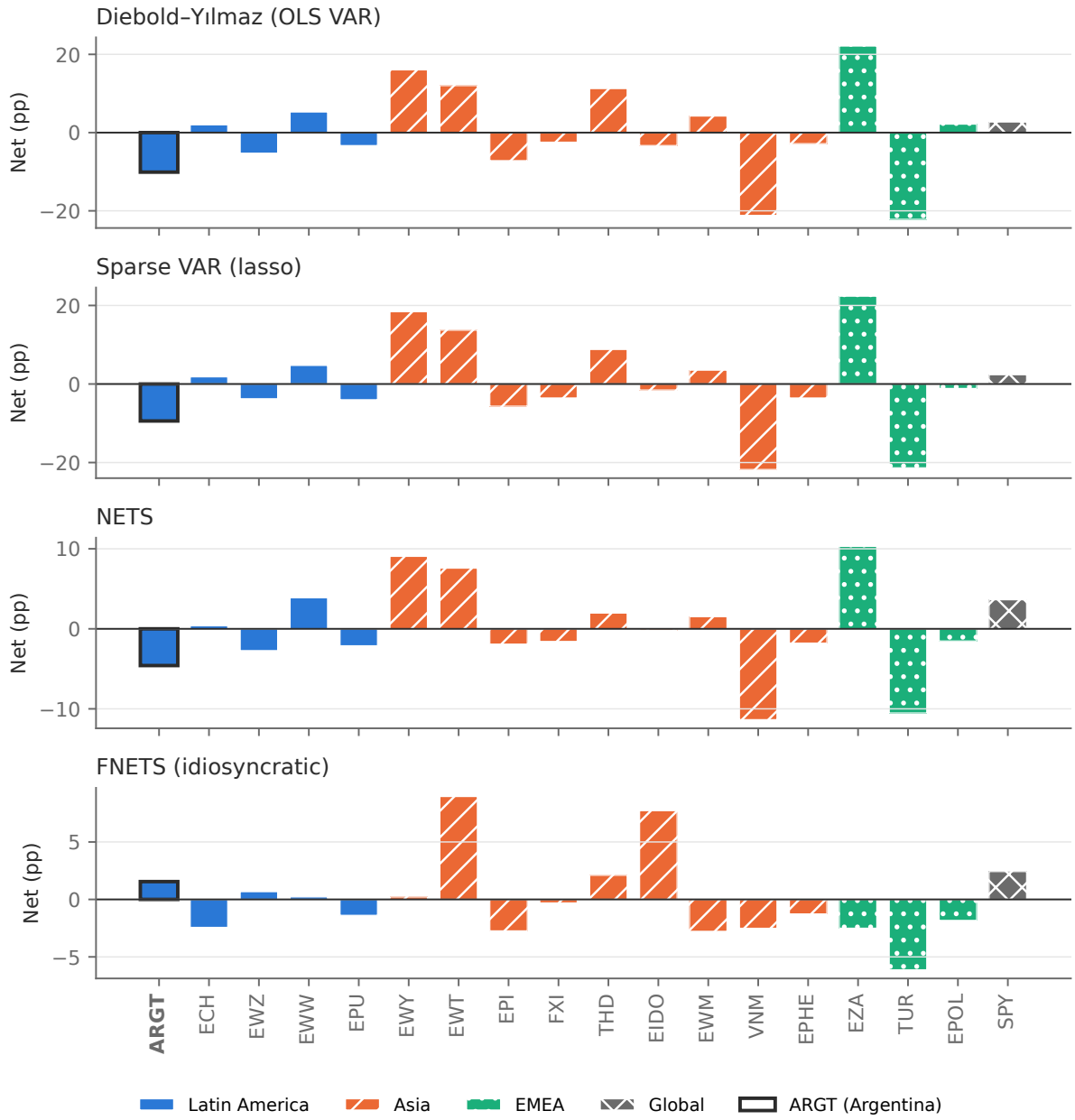


Fig. 5. Average net connectedness by country (transmitters > 0, receivers < 0), in percentage points. ARGT outlined.

6 Discussion

How much of measured connectedness is a common factor? A single factor explains on average 58% of the variance of weekly returns, and removing it lowers total connectedness by about 41% (38–43% across the robustness checks with one factor, 52% when two factors are imposed). This number, however, understates the role of the factor. Diebold–Yılmaz connectedness is a share of the forecast-error variance of the returns, whereas FNETS connectedness is a share of the forecast-error variance of the idiosyncratic component only, which is itself about 42% of the total. Computed market by market and window by window as (1 − common share) × idiosyncratic "from" connectedness (Section 4.5), idiosyncratic spillovers account on average for 19.6% of the variance of a market, against the 81.0% that Diebold–Yılmaz attributes to shocks in other markets; in the full sample the same quantity is 14.6% (95% block-bootstrap interval 13.3–18.3%, Table 6). Most of what the dense measure reports as connectedness is therefore exposure to the same shock rather than transmission between markets. This is consistent with a global financial cycle (Rey, 2013): the factor is not simply the US market, since excluding the S&P 500 leaves both the common share and the loss of connectedness unchanged, so it is a factor common to emerging markets as a group. It is also consistent with Forbes and Rigobon (2002). The peak of Diebold–Yılmaz connectedness in the windows containing March 2020 coincides with a peak of the common share, while idiosyncratic connectedness shows no peak at all: what looks like contagion in the dense measure is an increase in the weight of the common shock, together with higher volatility, and not an increase in bilateral links. The same reading applies to the decline after 2020, which comes with a smaller common share (58% to 47%) and a slightly higher idiosyncratic connectedness. The high correlation (0.95) between the common share and Diebold–Yılmaz connectedness is partly mechanical, because both increase whenever all pairwise correlations increase; it shows that the two quantities summarize the same feature of the data, not that one causes the other.

A simple benchmark shows how much connectedness a common factor produces on its own. Suppose that there are no links at all between markets and that a single factor explains 58% of the variance of each of the $N = 18$ series. All pairwise correlations then equal 0.58 and, since at the weekly frequency the generalized variance shares reduce to squared correlations normalized by row (see the next paragraph), total connectedness is $17 \times 0.58^2/(1 + 17 \times 0.58^2) \approx 85\%$. The empirical counterpart, the pure-factor benchmark of Section 4.5, averages 81.1% over windows, against 81.0% for Diebold–Yılmaz, with a correlation of 0.98 over time. A world without any contagion would therefore already produce as much connectedness as Diebold–Yılmaz measures in our data. The dense measure is thus insensitive to the idiosyncratic links that FNETS detects, which is consistent with the near-zero correlation between the two series (Section 5.4). Note also that, with row-normalized shares, total connectedness cannot exceed $(N - 1)/N \approx 94\%$ when $N = 18$, so the observed range of 72–89% lies close to its upper bound.

Does the method change the picture? Regularization alone does not. The lasso sets almost all lag coefficients to zero and still reproduces Diebold–Yılmaz connectedness (correlation 0.98), and the horizon has virtually no effect. Both facts have the same explanation: at the weekly frequency the moving-average coefficients Ψ_h for $h \geq 1$ are close to zero, so the generalized variance shares in equation (1) reduce to approximately ρ_{ij}^2, the squared correlations, normalized by row. In this setting Diebold–Yılmaz connectedness is close to a normalized squared-correlation matrix, the kind of pairwise and contemporaneous measure that Diebold and Yılmaz (2014) set out to improve on. The approximating model adds little dynamic content here because the data contain little. NETS keeps the same time pattern at a lower level because it regularizes the innovation concentration matrix, not only the lags, and sets weak partial correlations to zero; the connectedness it removes is mostly diffuse, small links. FNETS measures a different object, so the low overlap of its strongest links with those of the other methods is expected; that its connectedness is uncorrelated over time with theirs is instead an empirical

result (Section 5.4). The choice of method should therefore follow the purpose. For monitoring how tightly markets move together, Diebold–Yılmaz or the lasso VAR are preferable: they are smooth, robust to every specification we tried, and simple to compute. For portfolio diversification across emerging markets, the relevant quantity is the common-factor share, because that part of the co-movement cannot be diversified away within this universe. For the study of contagion, only a factor-adjusted measure such as FNETS addresses the right question, but in our setting it is also the least reliable estimate.

Transmitters and receivers. South Africa, Korea and Taiwan are net transmitters and Turkey, Vietnam and Argentina net receivers in the three systems estimated on returns, with a near-identical ranking (Spearman 0.96–0.99). These directions should not be read as evidence of lead–lag transmission. The Granger networks keep at most 2.5% of the possible lag links, so the data contain very little directional information; with negligible dynamics, the variance shares are close to symmetric squared correlations, and net connectedness mainly reflects the row normalization, that is, which markets are more or less correlated with the rest of the system. A market such as Turkey or Vietnam, weakly correlated with the others, mechanically appears as a receiver. Trading hours add a second source of bias, since all funds trade in New York after the Asian markets have closed. The idiosyncratic network gives a different ranking (Spearman 0.30 with Diebold–Yılmaz): Taiwan and Indonesia are the main transmitters and Argentina is close to neutral. Once the common factor is removed, Argentina neither transmits nor receives idiosyncratic shocks to a relevant extent, and its largest country-specific shock, the August 2019 primary elections, does not turn it into a relevant transmitter in any of the four systems: in our data, that local shock did not spread.

Limitations. First, we study returns, not the realized volatilities of the base paper, so our networks describe the transmission of return shocks, not of risk. Second, ETF prices embed exchange-rate movements against the dollar and US trading hours, which affect the estimated directions. Third, the rolling windows of 104 weekly observations are short for the sparse estimators, whose results depend on tuning choices: NETS jumps from one window to the next, and a finer grid of 60 penalty values does not remove this (mean absolute change of 4.5 against 4.6 points, Appendix A), so the jumps reflect the short windows rather than the grid; FNETS requires a positive-definite adjustment of the idiosyncratic covariance, gives an empty network in the full sample, and shows one-window drops that coincide with changes in the number of retained coefficients. With $N = 18$, the cross-section is also small for the factor-number criterion and for the consistency of the dynamic principal components, which rely on large N. Consistently, the loss of connectedness changes when two factors are imposed (52% against 41%, Appendix A). Fourth, almost no lagged links survive in the sparse methods, so the Granger networks carry little information. Fifth, consecutive windows share 100 of their 104 weeks, so the stability measures are high partly by construction; they are informative for comparing methods but not as absolute values. Finally, we report bootstrap confidence intervals only for the main full-sample quantities (Table 6), not for the averages over rolling windows, so small differences, such as 81.0% against 80.0% between Diebold–Yılmaz and the lasso VAR, should not be interpreted as significant.

7 Conclusion

Measured connectedness among emerging equity markets depends less on the estimation method than on whether the common factor is removed. At the weekly frequency connectedness is high, around 80% for the dense measures, and almost entirely contemporaneous; the lasso VAR and Diebold–Yılmaz agree closely, and NETS reproduces the same pattern at a lower level. A single common factor explains more than half of the variance of returns; spillovers between the

idiosyncratic components account for only about 20% of the variance of a market, against about 80% in the dense measures, and a common factor with no links between markets would by itself produce about 81% connectedness. Removing the factor lowers connectedness by about 40%, a share that is stable across windows, horizons and samples; the peak of 2020 and the decline afterwards reflect changes in the weight of this factor rather than in bilateral links. Most of the co-movement of emerging markets is therefore common exposure, not contagion. What cannot be claimed with the same confidence is the direction of spillovers between countries, which our data barely identify, and the structure of the idiosyncratic network, which is unstable in short windows. Natural extensions are to study volatility connectedness, as in the base paper, to use higher-frequency data or longer windows for the factor-adjusted estimators, and to add more markets, which would also improve the estimation of the common factor.

Statement on the use of AI tools

The authors used AI tools in the following ways during the preparation of this paper. Claude Deep Research (Anthropic) was used to search for and identify sources and data. Claude Code (Anthropic) was used to write the code that produced the estimations, figures and tables, and to typeset the paper in LaTeX. The research questions, the structure of the paper and the analysis and interpretation of the results are the work of the authors. The text of the paper, and in particular the research questions, the Discussion and the Conclusion, was written by the authors; Claude Code was used to convert this text into LaTeX format and for minor language editing. Claude (Anthropic) was also used to review a complete draft of the paper; the revisions that followed from that review (the idiosyncratic spillover share, the pure-factor benchmark, the additional robustness checks and bootstrap intervals, and the corresponding passages of text) were drafted with its assistance, implemented with Claude Code, and checked and approved by the authors. The authors reviewed all outputs and take full responsibility for the content of the paper.

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9 Data and Code Availability

The data (weekly ETF returns) and the full code are available at `https://github.com/ezellabra/ml4e-connectedness`. The pipeline is written in Python; NETS and FNETS are estimated in R (packages `nets` 0.9.1 and `fnets` 0.1.6) through a command-line bridge. All parameters are in `config.yaml` and the whole analysis is reproduced with `make all` (or `python run_all.py`).

A Robustness

Table 3. Robustness: mean total connectedness (%) by method, share of DY connectedness lost once the common factor is removed (FNETS), FNETS common-variance share, idiosyncratic spillover share of total variance, and pure-factor benchmark.

Scenario	Windows	DY	Lasso VAR	NETS	FNETS (idio.)	Lost w/o factor (%)	Common var. (%)	Idio. spill. (% total var.)	Pure-factor bench. (%)
Baseline (w=104, H=4)	179	81.0	80.0	61.5	47.2	41	58	19.6	81.1
H=2	179	80.9	80.0	61.5	47.2	41	58	19.6	81.1
H=8	179	81.0	80.1	61.5	47.2	41	58	19.6	81.1
w=78	185	81.2	79.6	57.8	50.5	38	58	21.0	80.6
w=130	172	81.1	80.5	63.4	46.2	43	58	19.3	81.6
Excluding SPY	179	80.0	78.9	59.9	45.9	42	58	19.0	80.2
Excluding windows touching 2020	141	79.7	78.6	58.8	47.2	41	55	20.8	79.7
NETS, 60-value grid	179	81.0	80.0	61.4	47.2	41	58	19.6	81.1
FNETS, q = 2 (fixed)	179	81.0	80.0	61.5	38.6	52	69	11.7	81.0

Table 3 reruns every method under alternative choices. The levels of Diebold–Yılmaz and lasso connectedness hardly move: between 79.7% and 81.2% and between 78.6% and 80.5% across all scenarios. NETS is the most sensitive to the window length (57.8% with 78-week windows, 63.4% with 130-week windows), as shorter windows leave the penalized estimator less information and make it keep fewer links. FNETS idiosyncratic connectedness ranges from 45.9% to 50.5%. The horizon matters very little: Diebold–Yılmaz changes by about 0.1 points between $H = 2$ and $H = 8$, and the sparse methods not at all, because their estimated lag coefficients are almost all zero and their variance decompositions are essentially driven by the contemporaneous

Table 4. Static connectedness: one estimate on the whole period (full sample) and on the subsamples before and after 2020. These are single long-sample estimates, not averages over rolling windows, so they differ from Tables 2 and 3.

Sample	Method	Total conn. (%)	LR p.corr. dens. (%)
Full sample	DY	81.9	100.0
Full sample	Lasso VAR	82.1	100.0
Full sample	NETS	73.5	60.1
Full sample	FNETS (idio.)	34.2	0.0
2011-2019	DY	83.0	100.0
2011-2019	Lasso VAR	82.8	100.0
2011-2019	NETS	69.6	60.1
2011-2019	FNETS (idio.)	34.1	2.0
2021-2026	DY	73.5	100.0
2021-2026	Lasso VAR	73.7	100.0
2021-2026	NETS	56.0	52.3
2021-2026	FNETS (idio.)	39.6	13.7

Table 5. Common-factor exposure and net connectedness by country: average FNETS common share over windows, average full-sample correlation with the other 17 series, and average net connectedness over windows (pp).

Ticker	Country	Common share (%)	Avg. corr.	Net DY (pp)	Net NETS (pp)	Net FNETS (pp)
ARGT	Argentina	48.6	0.47	-10.1	-4.6	+1.6
ECH	Chile	59.6	0.54	+2.0	+0.4	-2.4
EWZ	Brazil	55.2	0.52	-5.3	-2.7	+0.7
EWW	Mexico	61.9	0.57	+5.3	+3.9	+0.2
EPU	Peru	55.3	0.52	-3.3	-2.1	-1.4
EWY	South Korea	71.9	0.57	+16.0	+9.1	+0.2
EWT	Taiwan	68.7	0.57	+12.0	+7.6	+9.0
EPI	India	54.9	0.52	-7.2	-1.9	-2.7
FXI	China	56.4	0.49	-2.4	-1.6	-0.3
THD	Thailand	64.8	0.55	+11.2	+2.0	+2.1
EIDO	Indonesia	58.0	0.53	-3.3	-0.2	+7.7
EWM	Malaysia	62.1	0.55	+4.2	+1.5	-2.8
VNM	Vietnam	41.9	0.41	-21.2	-11.3	-2.5
EPHE	Philippines	56.8	0.53	-2.8	-1.8	-1.3
EZA	South Africa	74.4	0.61	+22.1	+10.3	-2.5
TUR	Turkey	35.0	0.39	-22.2	-10.5	-6.1
EPOL	Poland	58.3	0.53	+2.2	-1.5	-1.8
SPY	United States	61.3	0.56	+2.7	+3.6	+2.4

covariance. Excluding the S&P 500 lowers Diebold–Yılmaz connectedness by one point, and removing every window that contains a week of 2020 (141 windows left) lowers Diebold–Yılmaz, lasso and NETS connectedness by one to three points and leaves FNETS unchanged, without changing the ranking of the methods.

The main quantities are stable across scenarios. The share of Diebold–Yılmaz connectedness lost once the common factor is removed lies between 38% (78-week windows) and 43% (130-week windows), against 41% in the baseline, in every scenario with one factor; the variance share of the common factor is 55–58% with one factor. The ordering of the comparison criteria is also unchanged: the correlation of total connectedness with Diebold–Yılmaz is at least 0.96 for the lasso and between 0.84 and 0.90 for NETS, and in every scenario the dense estimators are the most stable from one window to the next, followed by NETS and then FNETS. With a finer grid of 60 penalty values, NETS total connectedness averages 61.4% (baseline 61.5%) and its mean absolute change between consecutive windows is 4.5 points (baseline 4.6). Fixing the number of factors at $q = 2$ gives a common share of 68.9% and a loss of connectedness of 52.2%, against 58% and 41% with the single factor selected by the criterion. The size of the loss therefore depends on the number of factors, and the 41% obtained with one factor should be read as a lower bound.

The subsample estimates in Table 4 show a lower level of connectedness after 2020 than before: Diebold–Yılmaz connectedness falls from 83.0% in 2011–2019 to 73.5% in 2021–2026,

Table 6. Full-sample estimates with 95% moving-block bootstrap intervals (blocks of 8 weeks, B = 200 samples; every estimation step is repeated in each sample).

Quantity	Estimate (%)	2.5%	97.5%
DY total connectedness	81.9	78.4	84.8
FNETS idiosyncratic connectedness	34.2	32.4	40.4
Common share	56.4	51.5	62.2
Idiosyncratic spillover share of total variance	14.6	13.3	18.3
Pure-factor benchmark	83.4	80.0	86.0

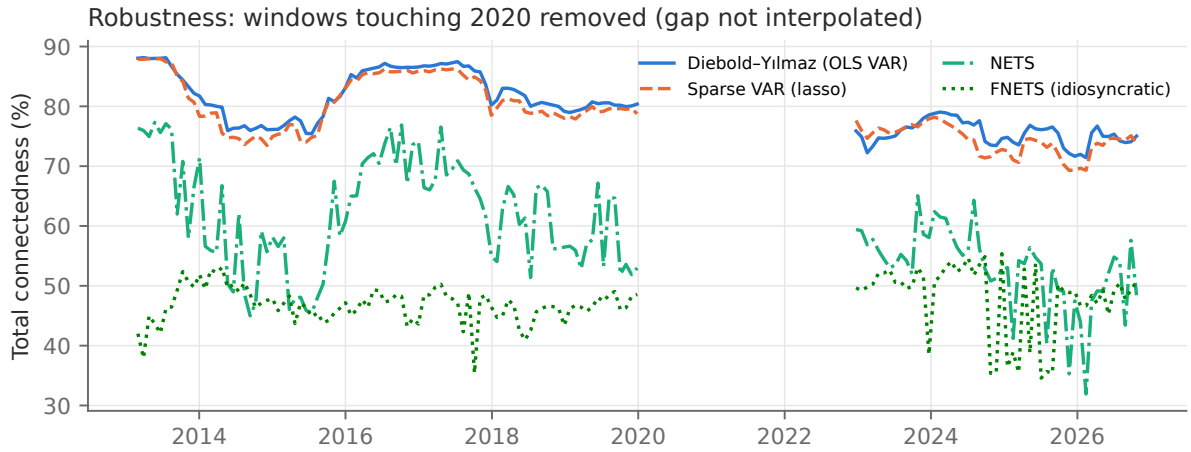


Fig. 6. Total connectedness excluding every window that contains a week of 2020.

and NETS from 69.6% to 56.0%. Idiosyncratic FNETS connectedness rises slightly (from 34.1% to 39.6%), while the variance share of the common factor falls from 58% to 47%. The decline after 2020 is therefore consistent with a weaker common component rather than with fewer idiosyncratic links. Figure 6 shows that, outside the windows that contain 2020, the time pattern of each method is the same as in the baseline.